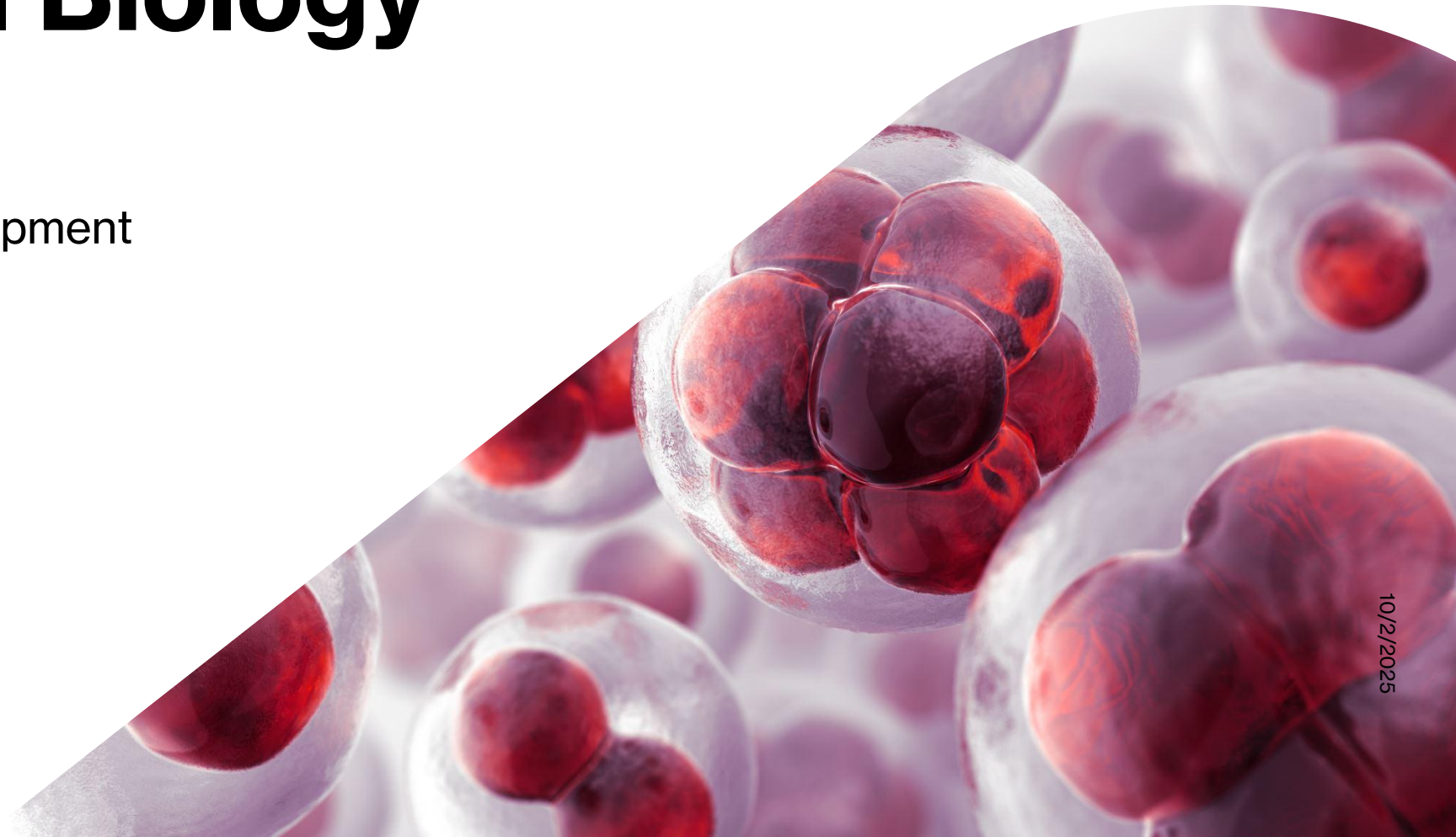
Stem Cell Biology

Section 2

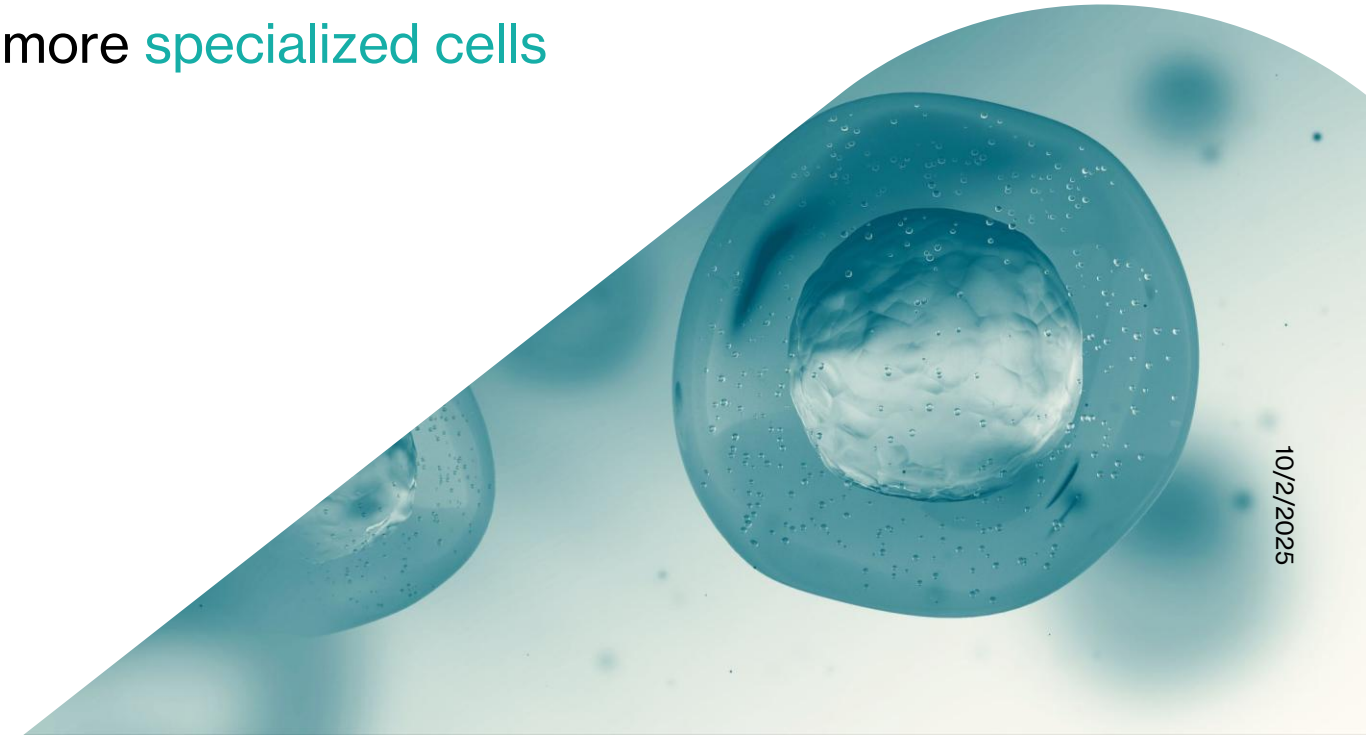
Early Embryonic Development



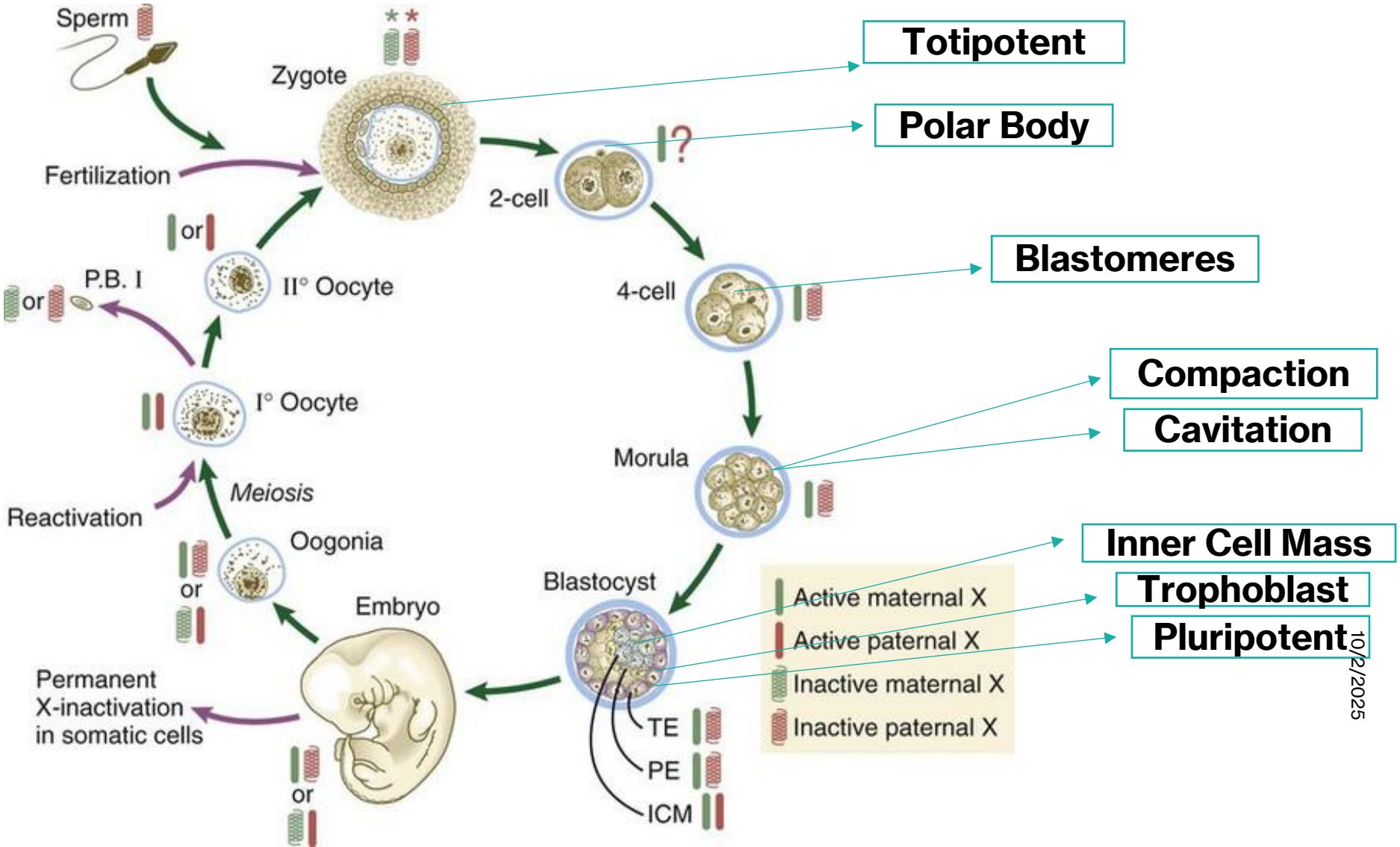
10/2/2025

Stem Cells

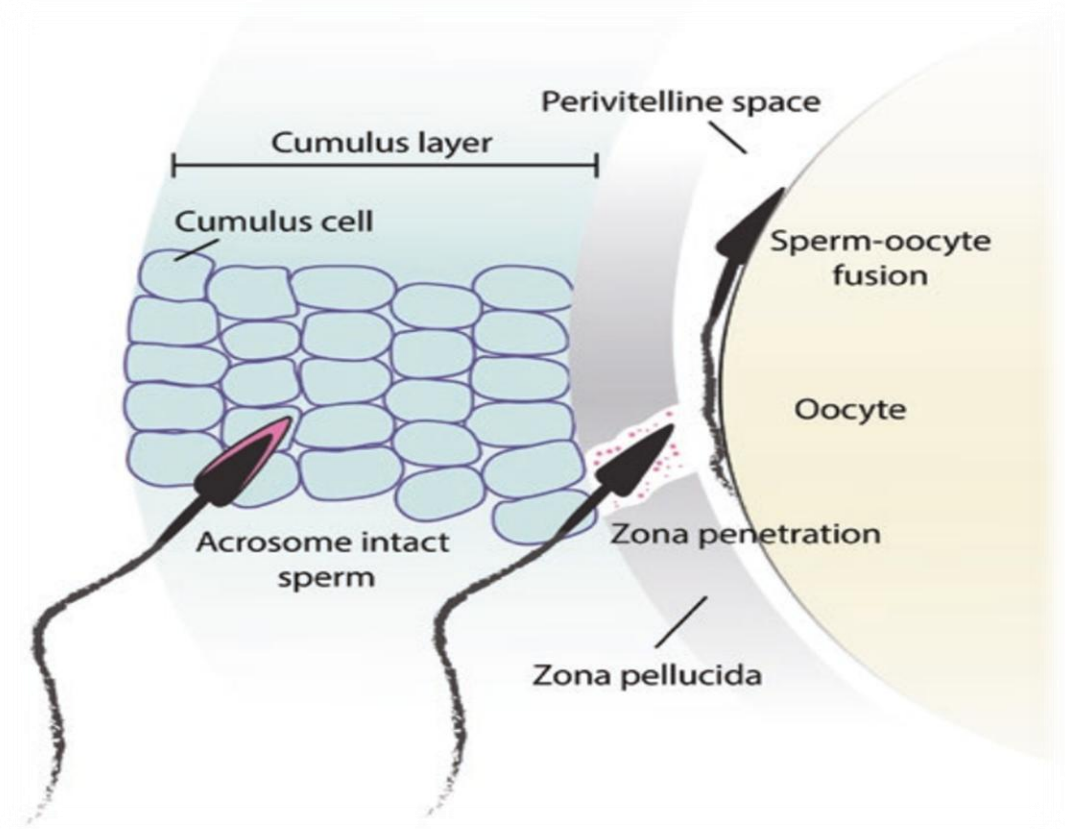
- **Stem cells are defined by two characteristics:**
- They can make copies of themselves, or self-renew.
- They can differentiate, or develop into more specialized cells

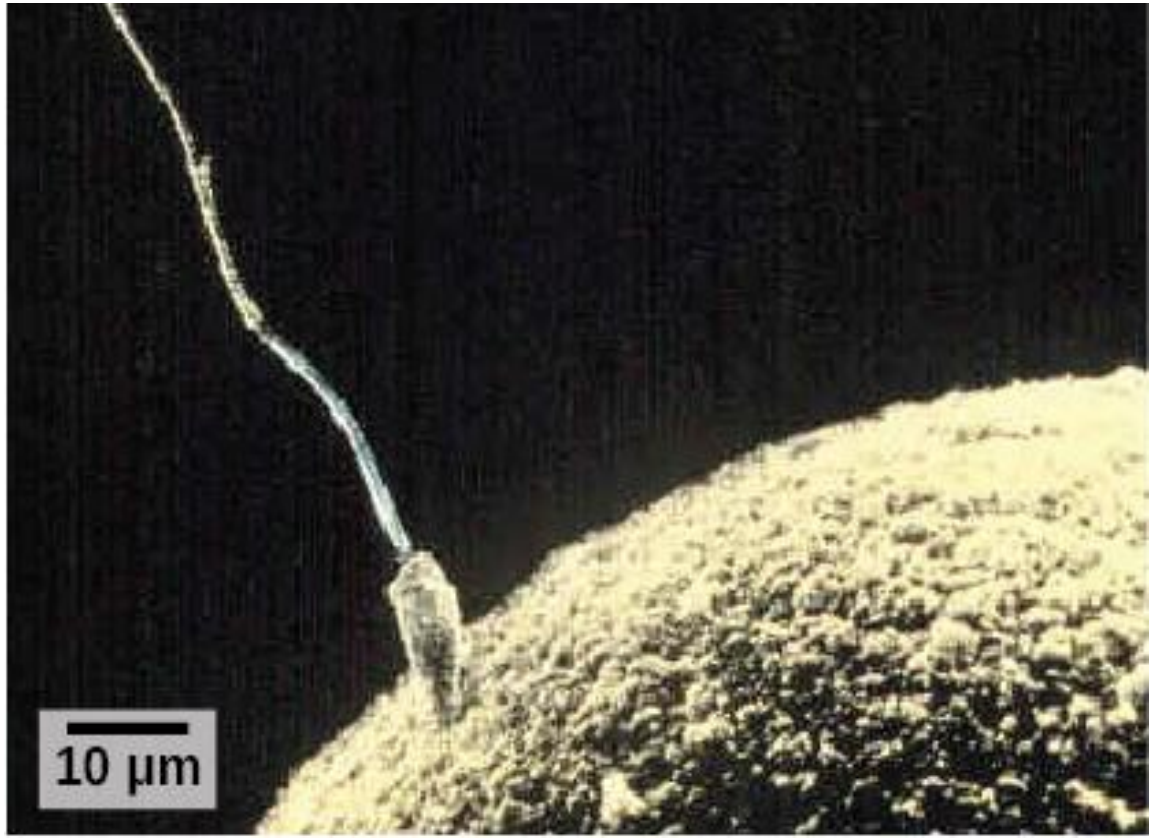


Stages of embryonic development showing embryonic stem cells

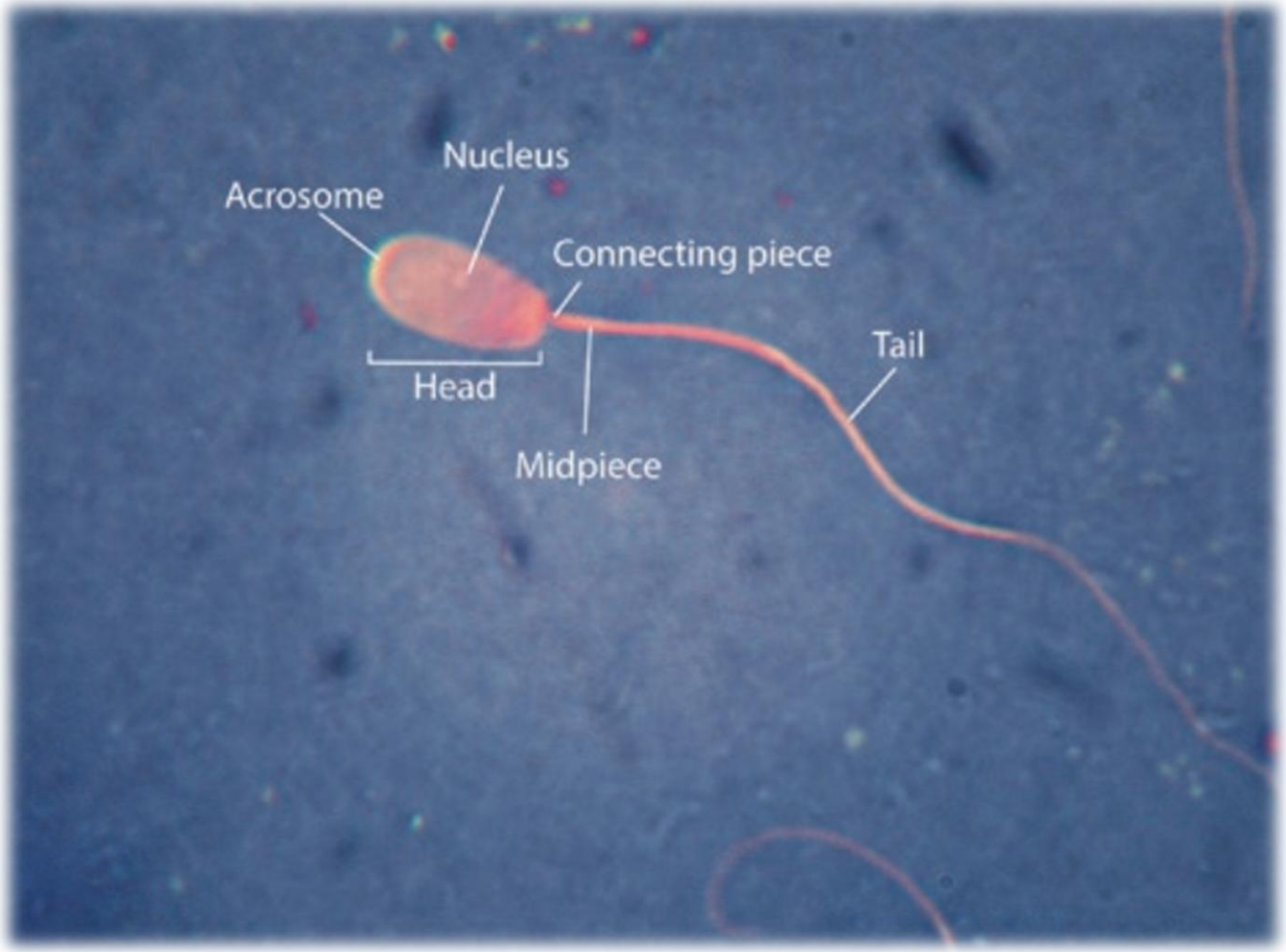


1- Fertilization (Day 1-2)



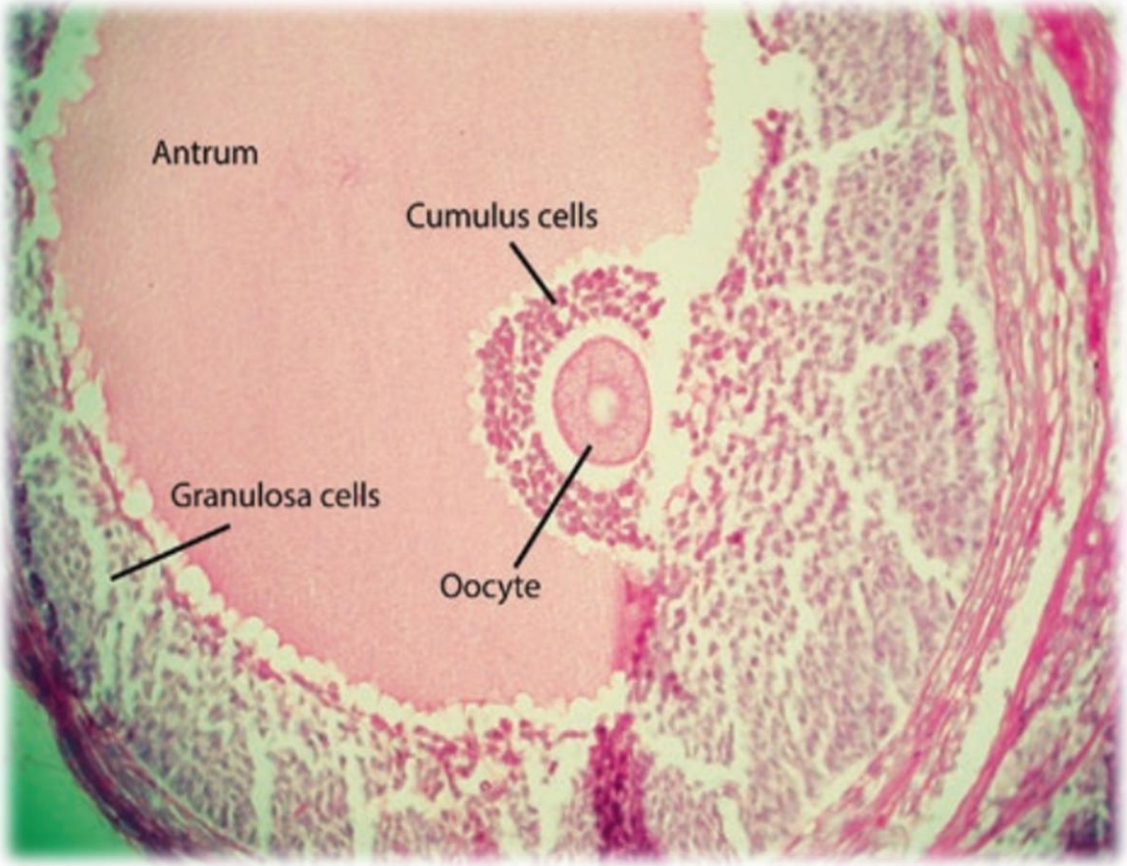


Sperm Morphology



Microscopic image

Oocyte Morphology





counterstained with hematoxylin

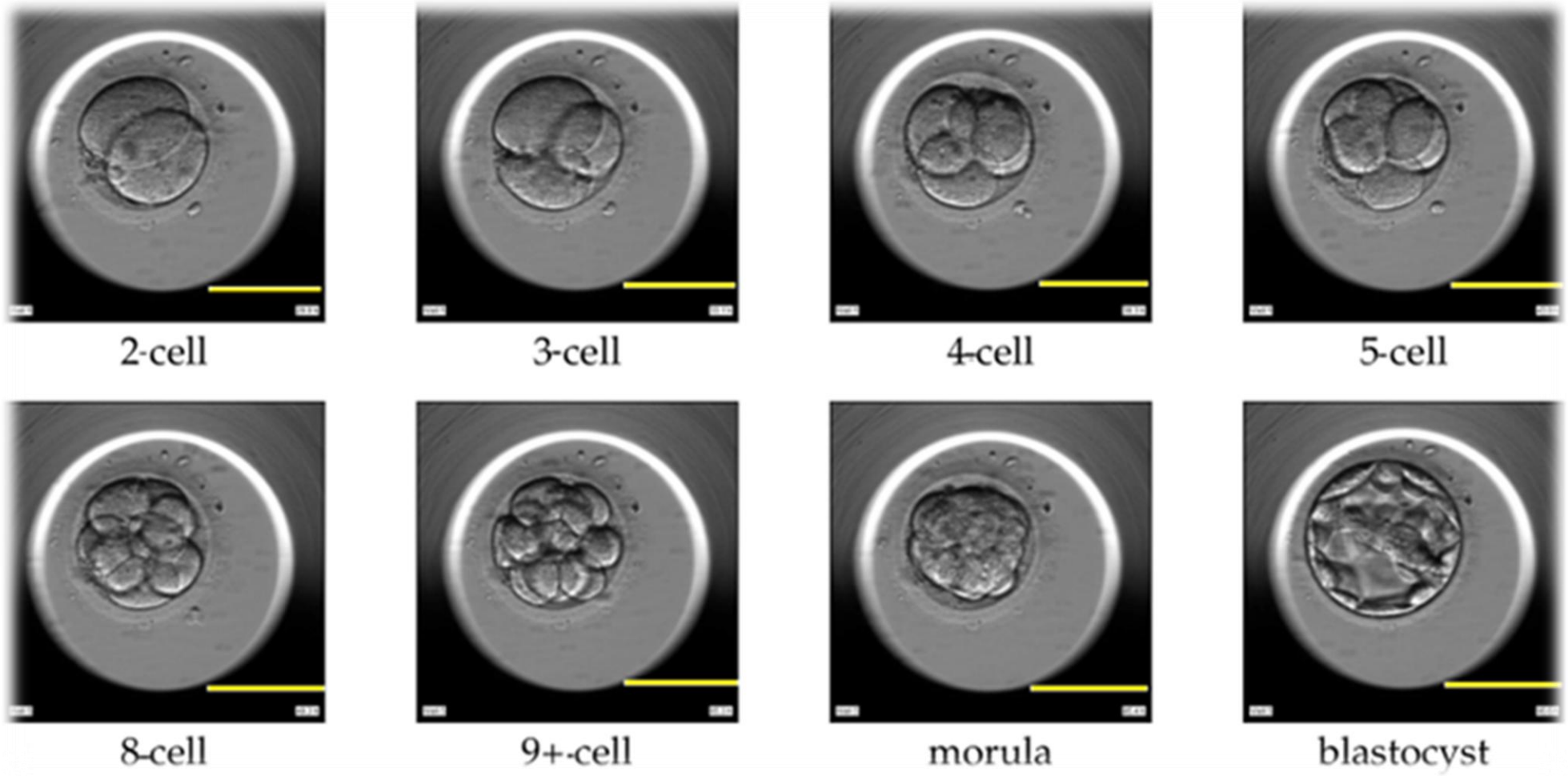
Zygote

A **zygote** is the first diploid cell that is formed by the fusion of male and female gametes resulting in the formation of an embryo.(**Totipotent**)

The **arrow** indicates the two circles (**pronuclei**) containing small dots (**nucleoli**), One of the pronuclei contains genetic information from the egg the other from the sperm.

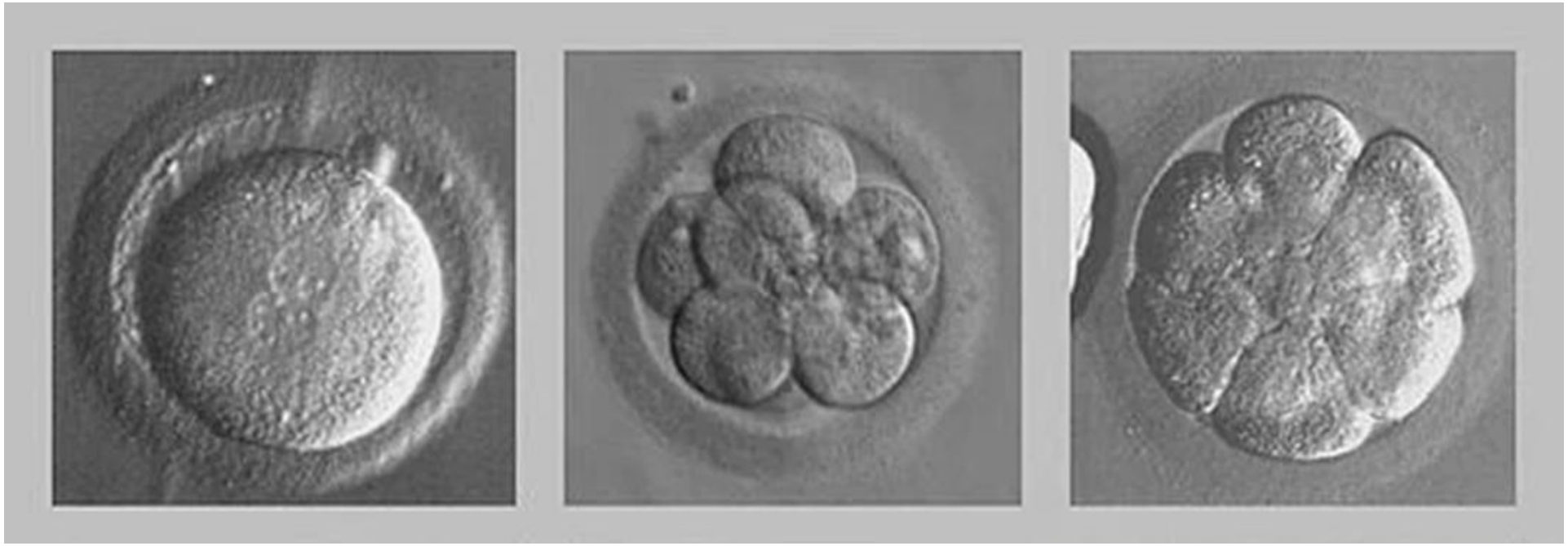


2- Cleavage (Day 3)



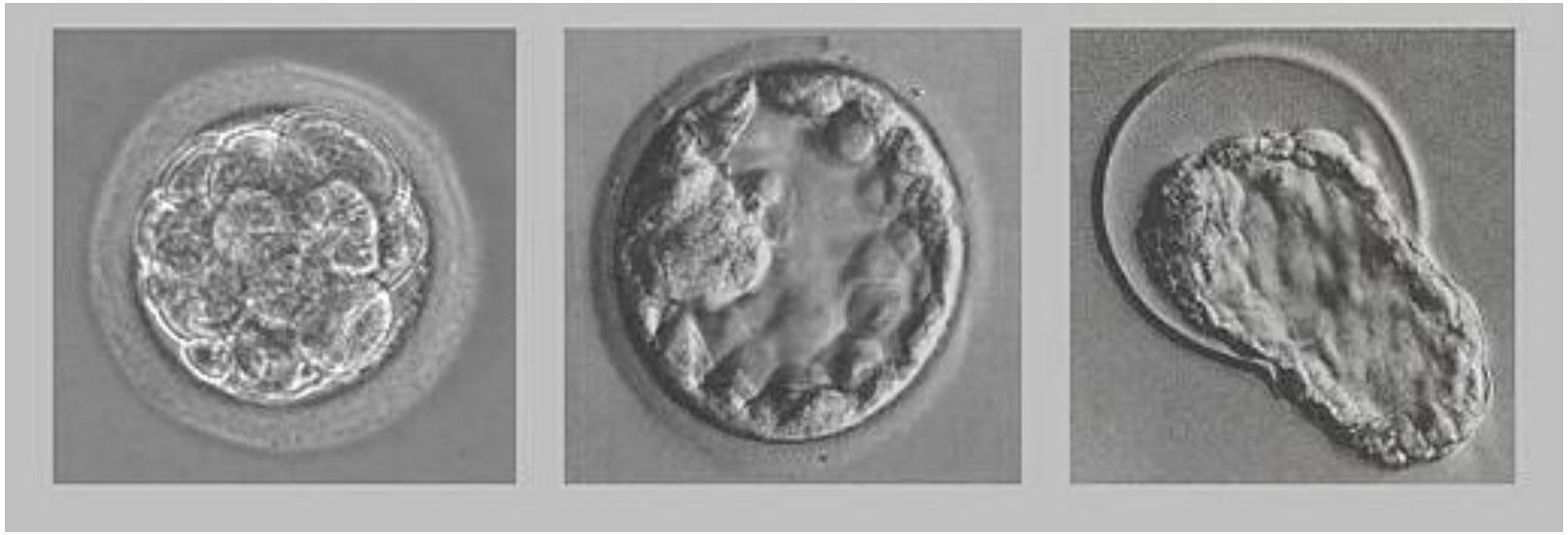
The large zygote is cleavaged into many smaller daughter cells called blastomeres.

2- Cleavage (Day 3)



From (left to right) showing Fertilized egg, 8-cell stage, cell adhesion

3- Morula Stage (Day 4)



compacted morula, a blastocyst, and zona hatching

3- Morula Stage (Day 4)



It consists of sixteen to thirty-two cells.(**Totipotent**)
The morula stage is the final stage prior to the formation of a fluid-filled cavity called the blastocoel cavity.

4- Blastocyst (Day 5)



Trophoblast
(Nourishes embryo)
(Become Chorion Later)

Blastocoel
(fluid-filled cavity)

zona pellucida

Blastocyst (70-100 cells) consists of two types of cells:
an outer epithelial layer (the **trophoblast**) that surrounds a small inner group of cells called the **inner cell mass (Pluripotent)** *(arrow)* .

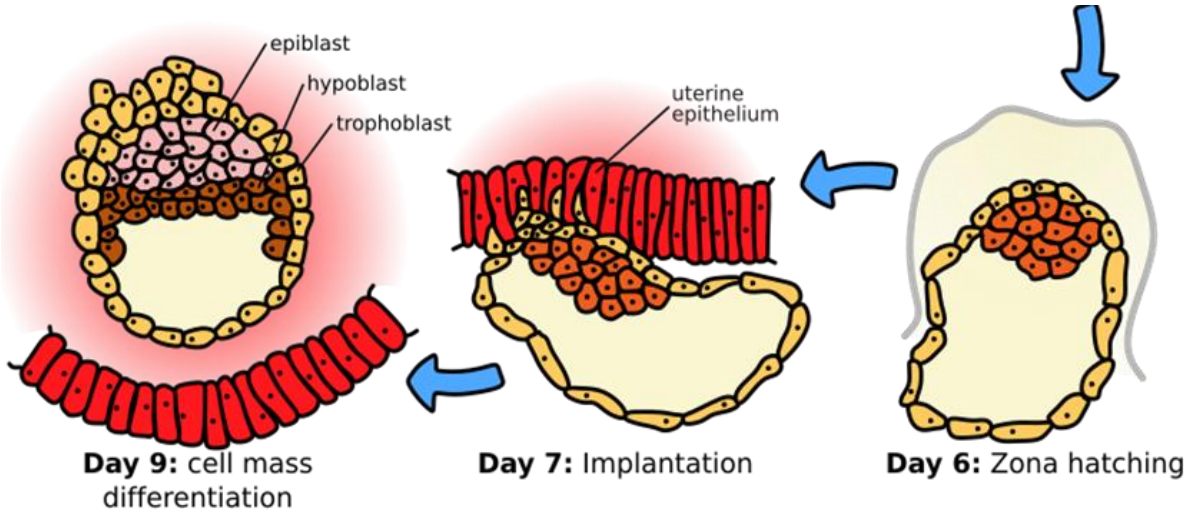
5- Zone Hatching (Day 6)

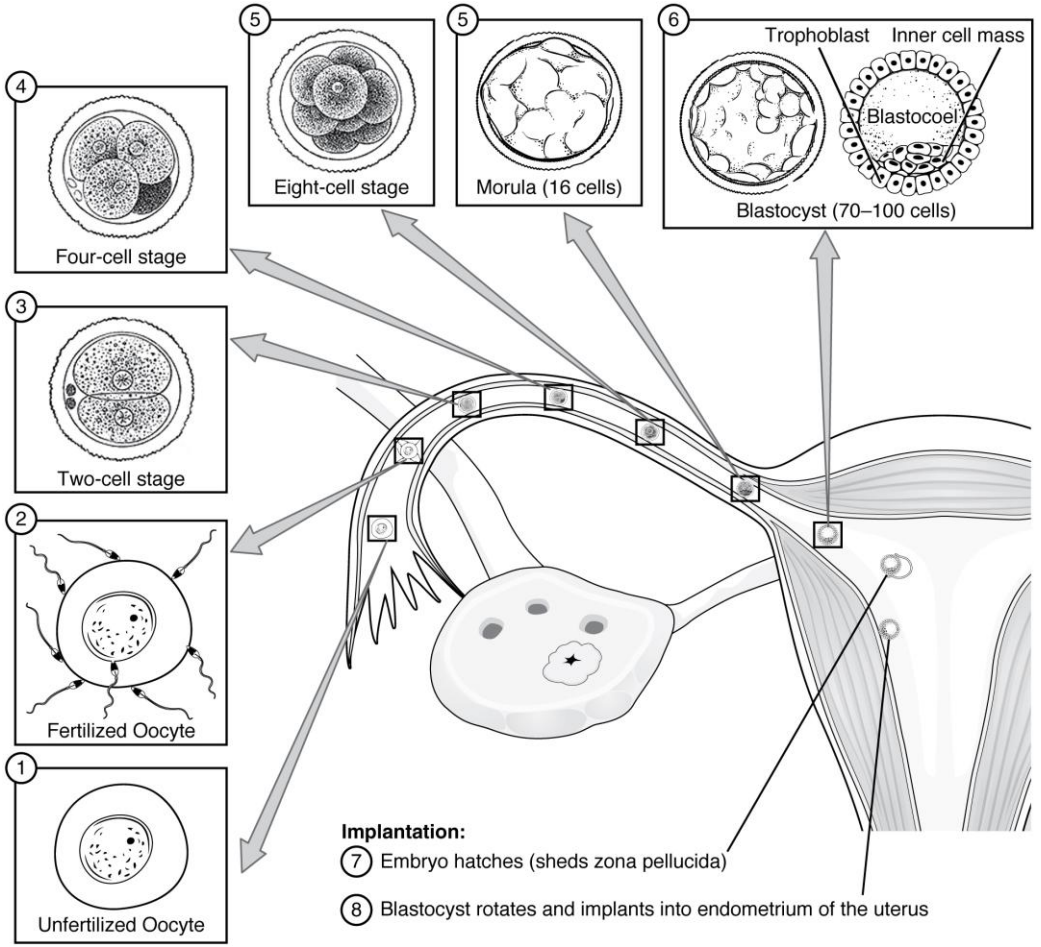


A hatching blastocyst beginning to protrude through the zona pellucida.

During blastocyst formation, the zona pellucida begins to disappear, allowing the ball of cells to proliferate, differentiate, change shape, and eventually implant into the uterine wall.

6- Implantation (Day 6)



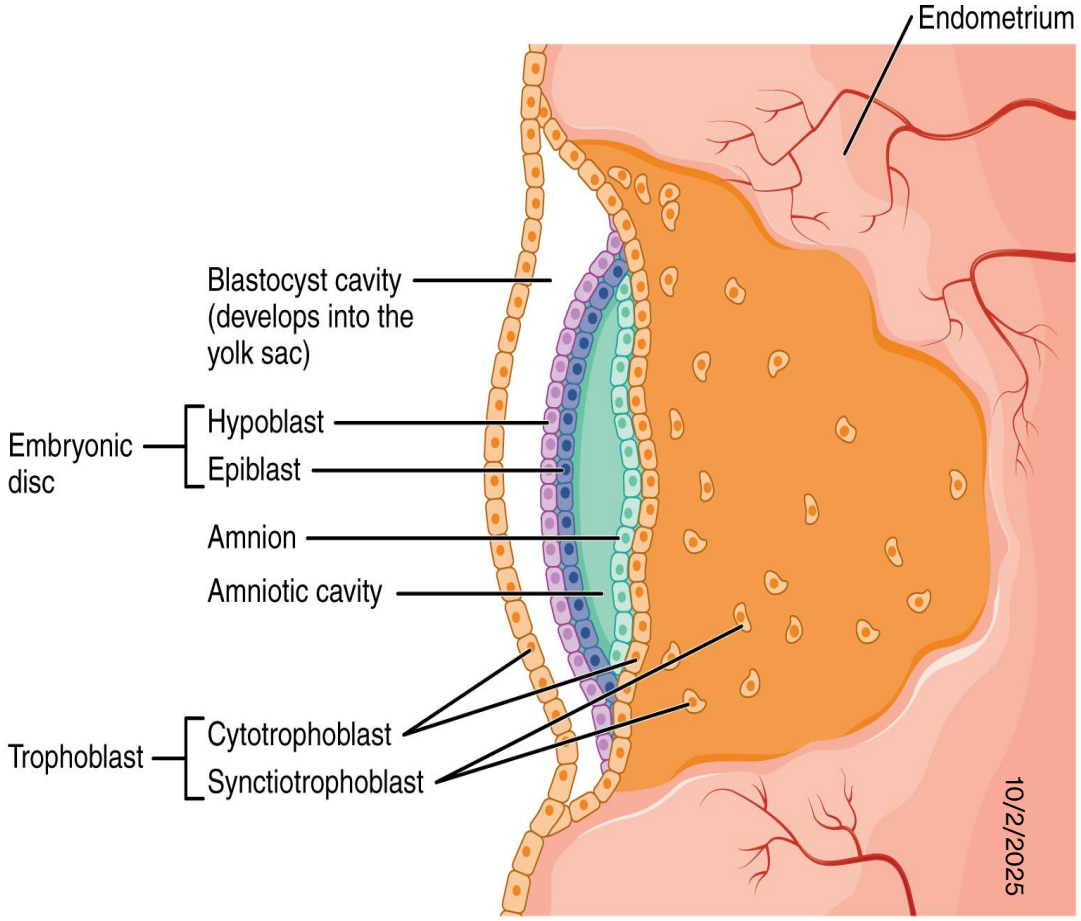


The blastocyst hatches, attaches, adheres, and invades into the endometrium of the female's uterus.

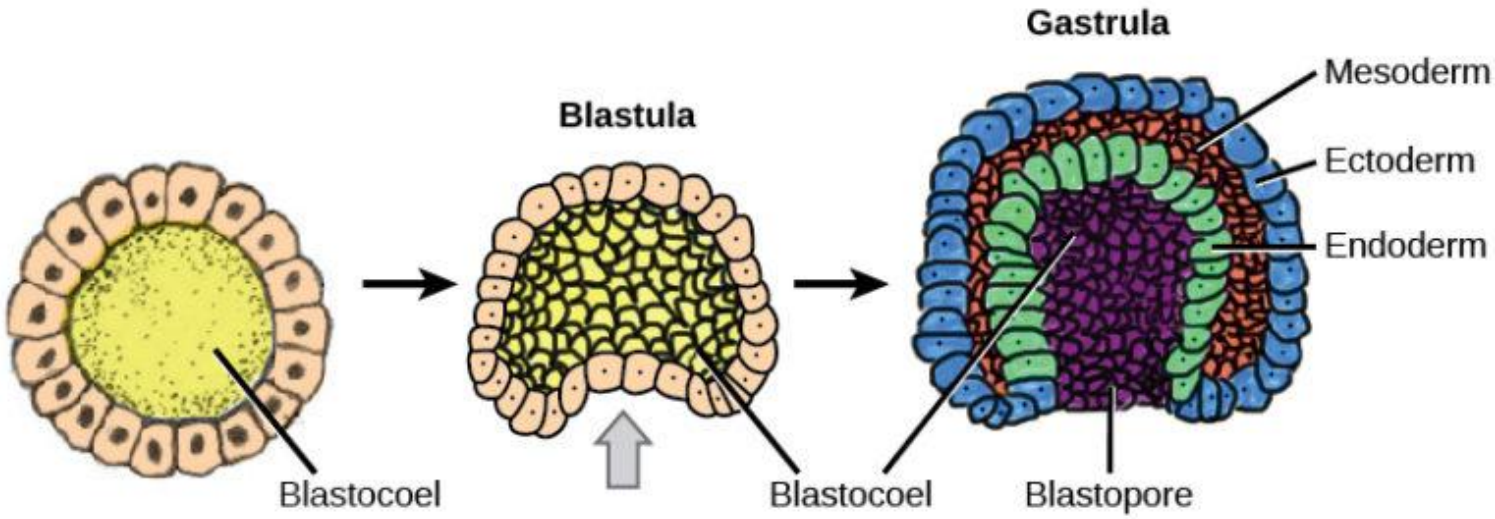
The superficial cells of the trophoblast fuse with each other, forming the **syncytiotrophoblast**,

Embryonic Membranes (At 2nd week)

- The cells of the inner cell mass form into a **two-layered disc** of embryonic cells, and a space.
- **The amniotic cavity** – opens up between inner cell mass and the trophoblast.
- Cells from the upper layer of the disc (the **epiblast**) = Form Amnion
- Cells in the lower layer of the embryonic disk (the **hypoblast**)= Form Yolk Sac
- The last of the extra-embryonic membranes is the **chorion**, which is the one membrane that surrounds all others.

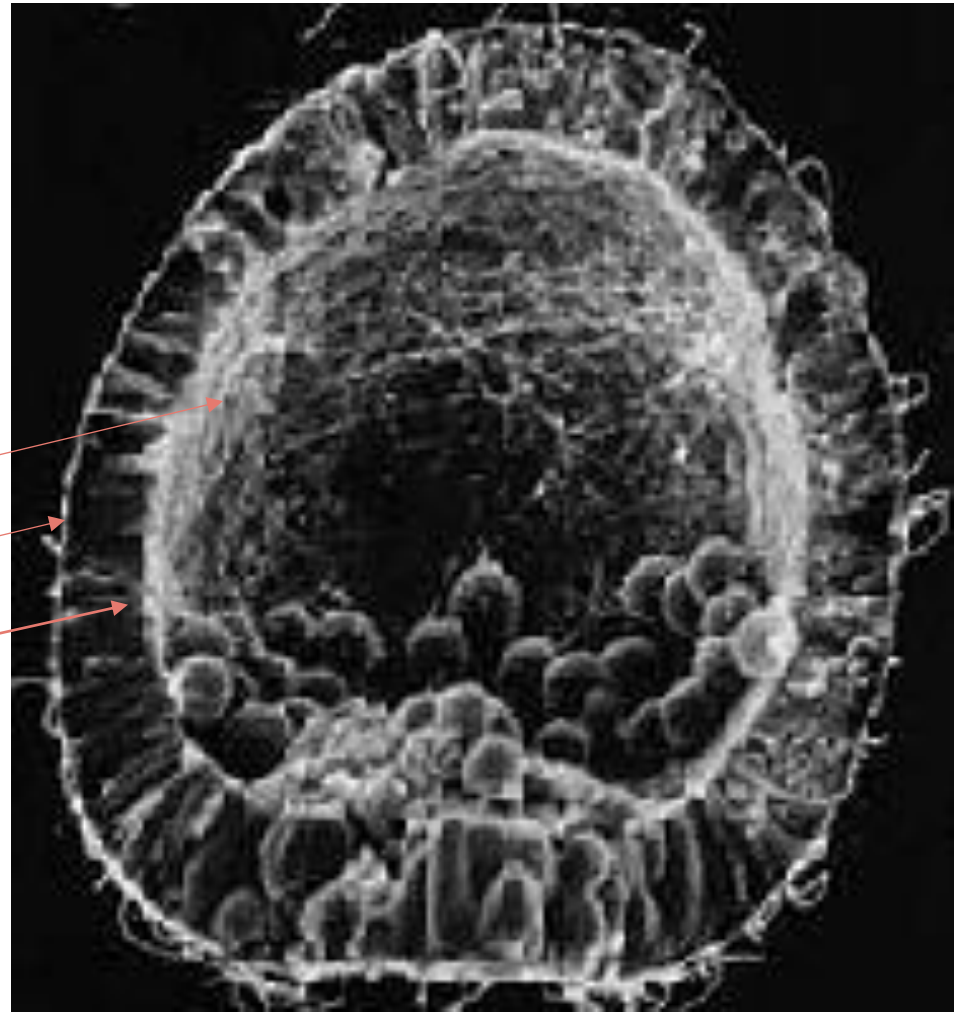


7- Gastrula Formation (At 3rd week)

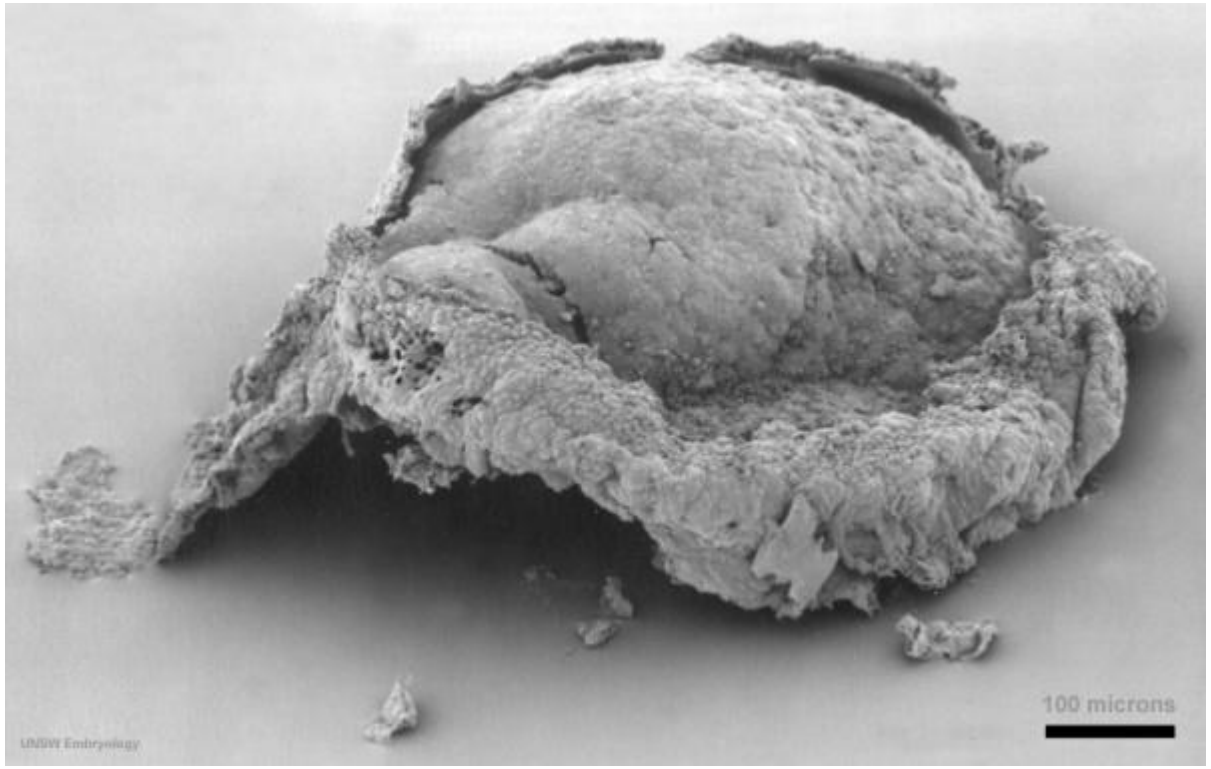


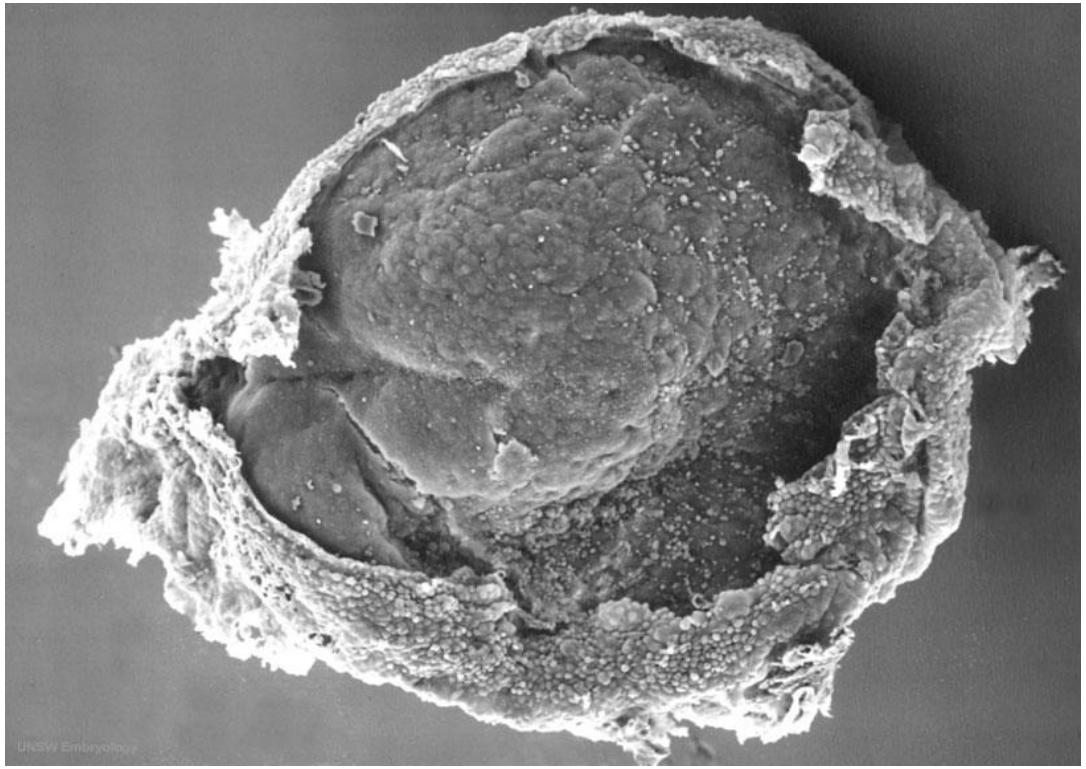
Gastrulation attains a three tissue-layered organism composed of endoderm, mesoderm, and ectoderm **(cells transition from totipotency to multipotency)**.

Gastrulation

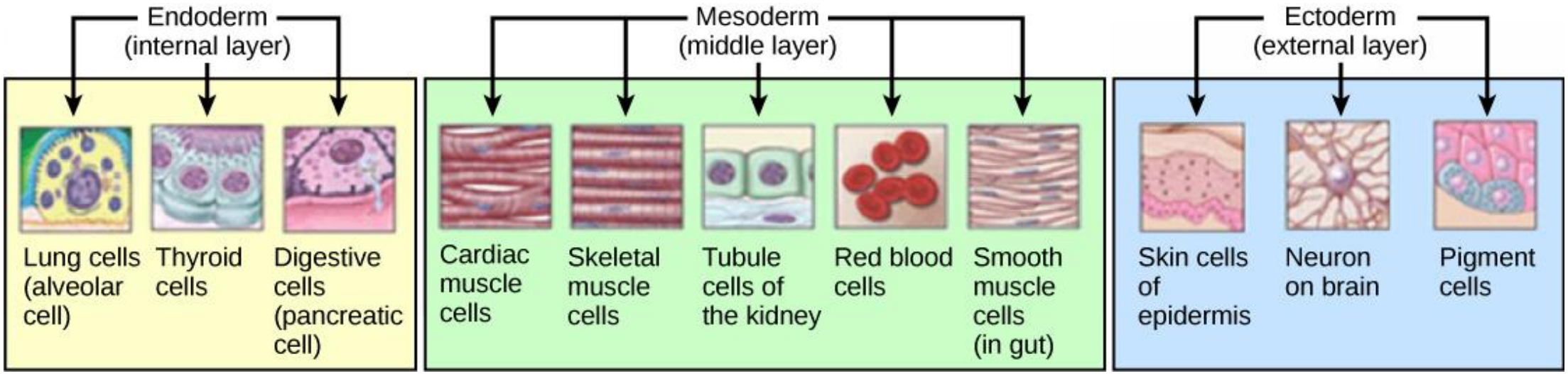


Gastrulation (Scanning EM)

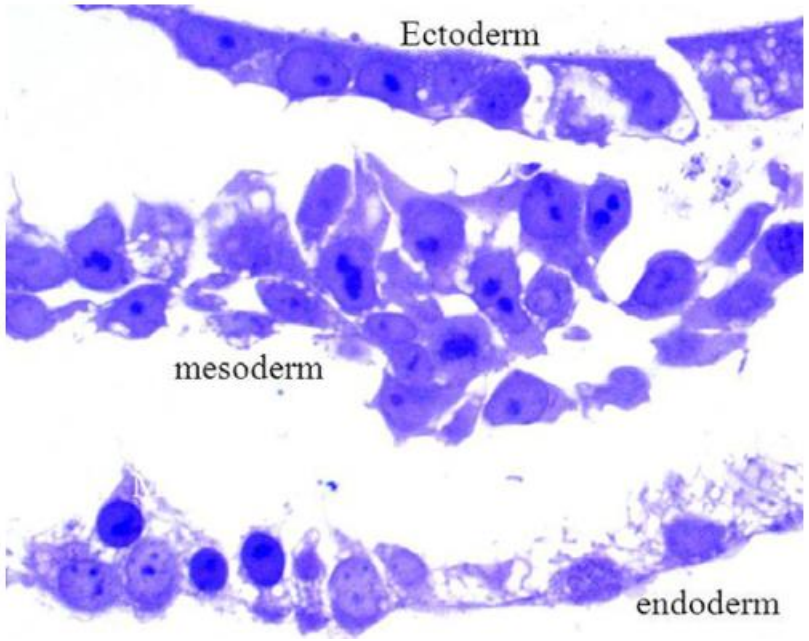


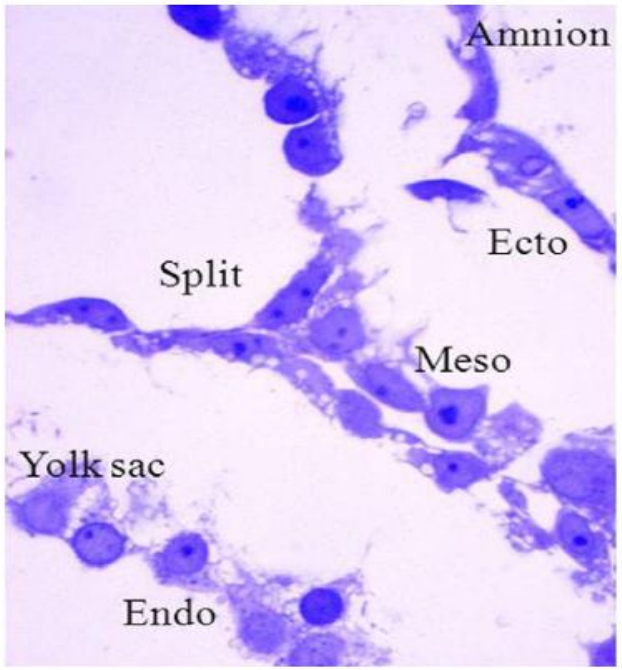


Germ Layers in Embryo:

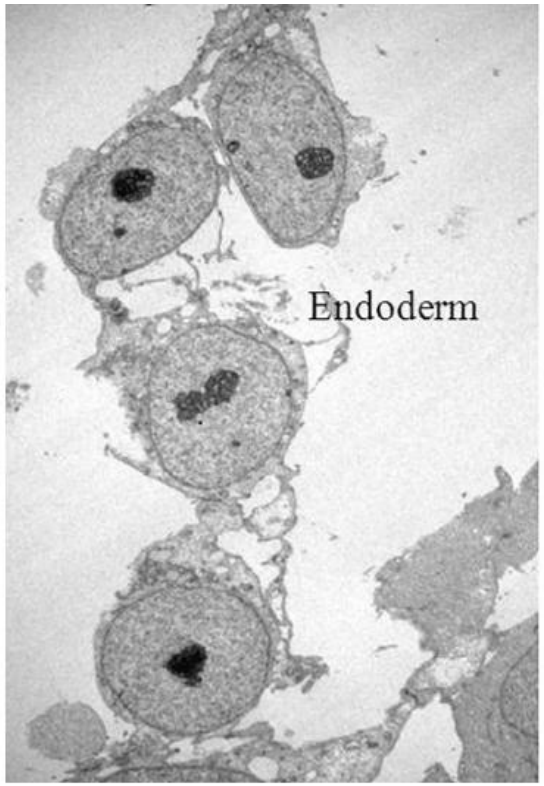


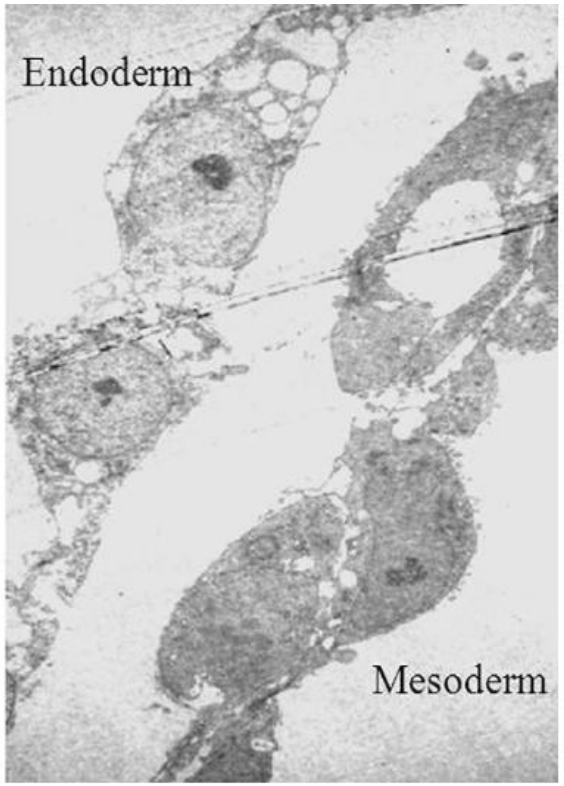
3 Germ Layers (Light microscopy)

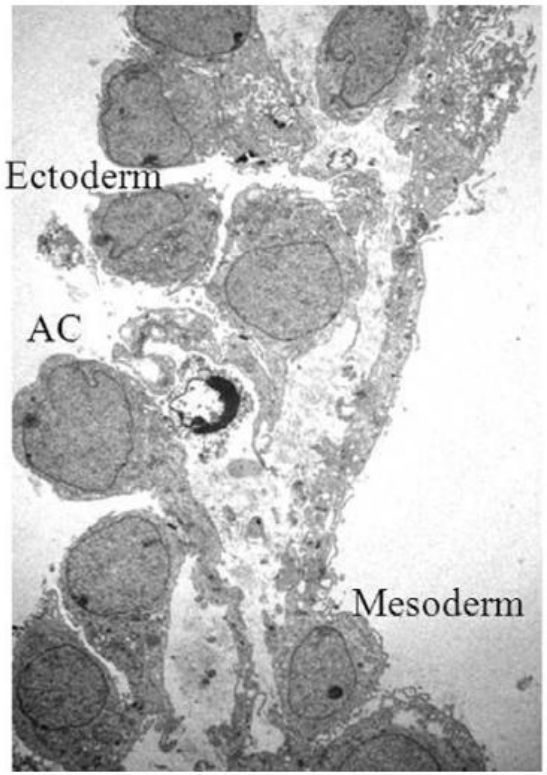


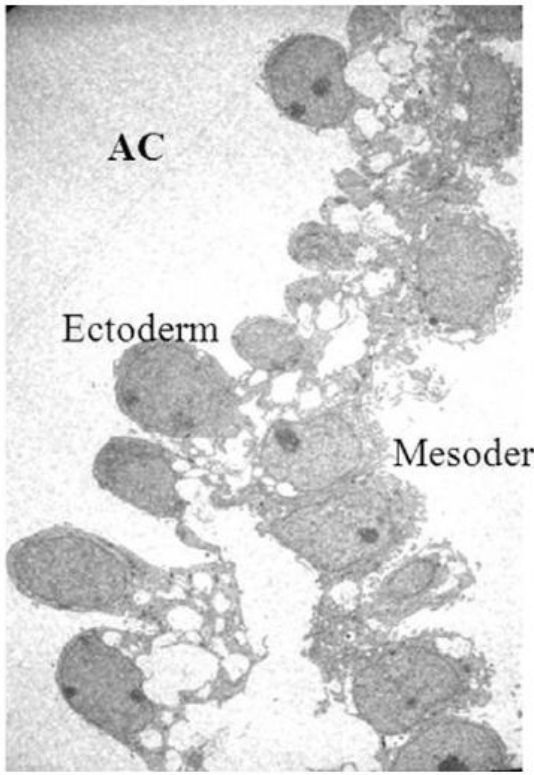


3 Germ Layers (Transmission electron microscopy)



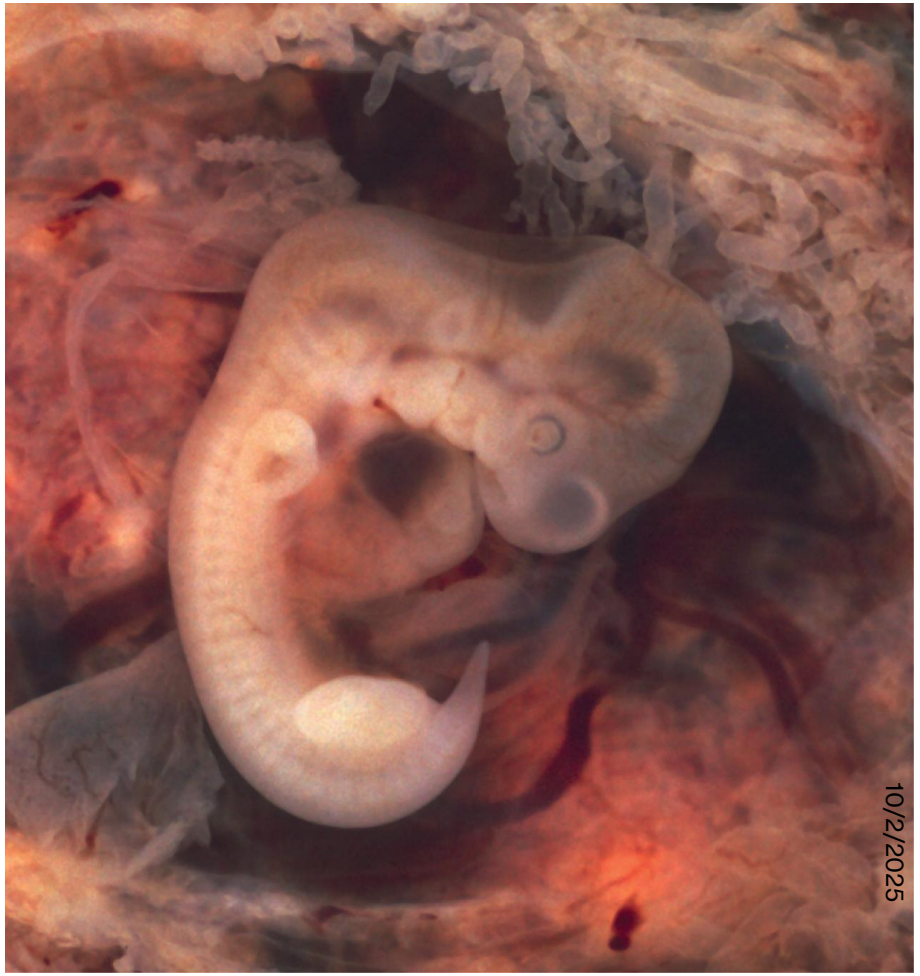






Organogenesis

- Within the first 8 weeks of gestation, a developing embryo establishes the rudimentary structures of all of its organs and tissues from the ectoderm, mesoderm, and endoderm. This process is called **organogenesis**.



Embryo at 7 Weeks